Practical work 6
Analyse & compare clustering models

Dataset (see attached files)

- Iris dataset:
  - https://archive.ics.uci.edu/dataset/53/iris
- Spotify dataset
  - https://www.kaggle.com/code/vikarna/spotify-music-data-analysis

Goal

Develop a machine learning model for clustering using algorithms such as:

- K-Means (for numerical data)
- PAM (Partitioning around Medoids)
- CLARANS (Clustering Large Applications based on Randomized Search)
- CLARA (Clustering Large Applications)
- Hierarchical (Agglomerative and Divisive methods)
- Density-based method (DBSCAN)
- Model based (Self-Organizing Map (SOM))

Implementation and Results

- Implement the selected clustering algorithms for clustering.
- Adjust parameters and compare the clustering performance.

Performance Evaluation & Visualization

- Evaluate the clustering performance using the metrics: inertia, silhouette, calinski harabasz, davies_bouldin.
- Visualize the model's effectiveness of the clustering algorithm by plotting the optimal k .
- Interpret the results to identify key features and best K values.

Conclusions

- Benchmark the clustering performances with other clustering methods.
- Provide recommendations for optimizing K values of clusters.